A Correlational Study of Emotional Intelligence, Cognitive Flexibility and Psychological Performance in Sports

Satya Narayan Tiwari[1], Deepak Mishra[2], Meenal Gupta[3], Sarika Alreja[4]

ABSTRACT

In recent years, there has been a lot of attention turned towards understanding and managing the relationship among affective and cognitive processes and sport performance. Therefore, the present study was planned to investigate the role of emotional intelligence in predicting sports performance of combat (Boxing, Fencing & Wushu) and non-combat (Hockey & Cycling) sports and how cognitive flexibility mediates the relationship between these. A Total of 179 athletes (101 from combat sports and 78 from non-combat sports) including male and female from STC schemes of SAI, NS NIS, Patiala, were selected based on purposive random sampling method and participated in the study. Their age range was between 15-25 years and minimum 10th or High school level of education was ensured. The sample consisted of participants from mostly middle socio economic status of semi urban back ground. After collecting demographic details of the athletes, The Multidimensional Self-Report Emotional Intelligence Scale Revised (MSREIS-R), Cognitive Flexibility Inventory (CFI) and Psychological Performance Inventory (PPI) were administered on the athletes and data was collected. Data thus obtained were analysed using chi-square, t-test and Pearson's Correlation through SPSS software. Results of the study show that Cognitive Flexibility was found to be significantly correlated with Psychological Performance for the whole sample as well as both the subgroups. Emotional Intelligence also had a positive but not statistically significant correlation with Psychological Performance. However, Emotional Intelligence was found to have a statistically significant positive correlation with Cognitive Flexibility for the Non-Combat sample. Hence, findings of the study underline that devising specific strategies for enhancing cognitive flexibility and emotional intelligence will further enhance the athletes' psychological performance and in turn the overall game performance.

Key Words:

Emotional Intelligence, Cognitive Flexibility, Psychological Performance

---

1. Senior Research Fellow, Sports Psychology, SAI NSNIS, Patiala, Punjab, Mail id: satyantiwari@gmail.com
2. Junior Scientific Officer, Sports Psychology, SAI NSNIS, Patiala, Punjab, Mail id: deepak2821@gmail.com
3. Mental Trainer, SAI NSNIS, Patiala, Punjab, Mail id: meenalmalhotra17@gmail.com
4. Clinical Psychologist, Patiala, Punjab. Mail id: sarikaalreja@gmail.com

INTRODUCTION

Mental and emotional fitness can play an important role to better performance. Psychology has collaborated the sports performance through the use and practices of psychological theories, rules, regulations and techniques such as psychotherapy, counseling, imagery, and various other psychological techniques. In recent years, there has been a lot of attention turned towards understanding and managing the relationship among affective and cognitive processes and sports performance. Literature has identified a group of six broad psychological areas linked to affective and cognitive performance i.e. emotion, motivation, arousal, and activation, concentration, attentional control, regulation, of stress and coping with adversity (Hardy et al, 1996).

The influence of emotions in sports is crucial to performance; although sports can be practiced for fun and enjoyment (Sit & Lindner, 2005). Athletes also experience anxiety and stress when they try to reach a high performance (e.g., Mellalieu et al, 2009). The pressure and the subsequent stress and emotions experienced by athletes are more intense when highly valued goals are at stake (Lazarus, 2000). EI is important for both coaches and athletes. For coaches, it has been positively associated with coaching efficacy (Thelwell et al, 2008), more specifically for motivation efficacy and character building. For athletes, higher Emotional Intelligence has been linked to higher performance in team sports, such as Cricket (Crombie & Noakes et al, 2009), Hockey (Perlini & Halverson, 2006), and Baseball (Zizzi & Hirschhorn et al, 2003). According to Zizzi and colleagues (2003), an athlete must recognize one's emotions, as well as teammates and opponents emotions, in order to perform well in team sports. At the individual level, higher emotional intelligence was found to be positively related to the use of psychological skills, such as imagery and self-talk (Lane et al, 2009).

According to development of different methods of exercise and skill execution techniques, the quality of exercise sessions and having high power, speed, flexibility and physical preparedness cannot singly guarantee the success of athletes and or teams, but development and improvement of mental preparedness and paying attention to factors such as self-confidence, self-management, adequate emotional energy, awareness of one's and others feelings and emotion management would lead to transfer of learnt skills from exercise sessions to competition time in a positive manner.

Along with affective skills, cognitive flexibility is also found to play an important role in sports performance. Individuals with high level of cognitive flexibility easily adapt to new situations (Anderson, 2002), can deal with stress (Altunkol, 2011) and their worries decrease while their adaptation increases (Oz, 2012). According to Diril (2011) there is a significant positive relationship between anger management and cognitive flexibility, while Bilgin (2009) reported that problem-solving skills significantly predicted cognitive flexibility. Thus, it becomes important to understand the relationship between emotional intelligence and sports performance and how cognitive flexibility can alter this relationship.

Coaches and trainers often mention the pre-eminence of creative thinking ability in sports. However, it is not clear how creative thinking can be improved (Memmert & Bertsch et al. 2010). Just as in more explicit cognitive processes, participation in various sports and physical activities can be valuable for the development of creativity (Abernethy & Cote et al. 2005). It is important that the range of environmental variability is wide and this is perhaps necessary for the development of creativity. It is important for players to have a wide range of experience and to improve their ability to cope with unexpected situations (Memmert & Bertsch et al. 2010). Especially in sports with tactical response patterns such as football, hockey or basketball, and offensive games where original solutions are critical.

A significant positive relationship was found between emotional intelligence and cognitive flexibility by Bulent (2013). In addition, it was also concluded that emotional intelligence and cognitive flexibility have significant negative relationships with anxiety and depression. It is concluded that high levels of emotional intelligence and cognitive flexibility are associated with psychological performances. On the other hand, the regression analysis shows that evaluating emotion, which is one of the subscales of emotional intelligence, is the strongest predictor of anxiety and depression. Cognitive flexibility is significantly related to psychological symptoms but makes little contribution to prediction. The significantly positive relationship between emotional intelligence and cognitive flexibility would be expected, and supports the findings of Davis and Humprey (2012); Sahin et al. (2009) and Gannon & Ranzijn (2005).

Hence, the present study was planned to explore the above said possibilities and find a relationship between emotional intelligence and cognitive flexibility that could predict the level of performance according to the type of game.

METHODOLOGY

Design:

It was a cross sectional between groups study carried out using correlation design.

Sample:

The sample for this study comprised of 179 participants of various games training at NSNIS, Patiala, under the STC Scheme of SAI. Out of the 179, 101 subjects were from combat sports and 78 subjects were from non-combat sports including both male and female athletes. Combat sports group comprised of athletes of Fencing, Wushu and Boxing while the non-combat sports group comprised of athletes of Hockey and Cycling. The sample selection was done using the purposive random sampling method. Most of the participants were in the age range of 10-20 years, educated up to a minimum of High School level, from middle socio-economic status of semi-urban background of Punjab and Haryana.

Tools

- Socio-demographic Data Sheet
- The Multidimensional Self-Report Emotional Intelligence Scale Revised (MSREIS-R: Pandey & Anand, 2008)
- Cognitive Flexibility Inventory (CFI: Dennis & Vander Wal, 2010)
- Psychological Performance Inventory (PPI: Loehr, 1986)

Procedure:

After getting informed consent from each athlete, they were requested to complete the socio-demographic profile, emotional intelligence questionnaire, cognitive flexibility inventory and psychological performance inventory. The data were collected at SAI, NS NIS Patiala campus. All of the study participants were inmates of the STC Scheme of SAI.

Statistical analyses:

The data obtained were compiled and statistically analyzed with the help of SPSS software. Chi-Square, t-test and Pearson's Correlation were calculated to establish the relation between selected variables as per the planned objectives of the study.

Results & Discussion

After analyzing the data using the SPSSS (IBM 20v version) software, the following results were obtained which are depicted in Table 1.

Table -1 : Showing Socio-demographic characteristics of the study sample

Socio- demographic Variable		Game		Chi-value	Sig.
		Combat	Non-combat		
Gender	Male	65	43	1.566	.211
	Female	36	35		
Education	Below ten	19	11		.094
	10th	39	40		
	12th	35	15	7.125	
	Graduation	8	12		
Religion	Hindu	68	45		.416
	Sikh	32	32	1.755	
	Muslim	1	1		
Family income	<0.5 L	15	17		
	0.5-1 L	22	20		
	1-5 L	44	26		
	5-10 L	17	12	2.802	.592
	>10 L	3	3		
Residential Background	Rural	36	33		
	Urban	57	37	1.454	.483
	Semi urban	8	8		
Family Structure	Joint	31	23	.030	.862
	Nuclear	70	55		
Father Education	<10th	14	9		
	10th	19	14		
	12th	37	31	1.268	.867
	Graduation	18	17		
	PG	13	7		

**Significant at .01 level *Significant at .05 level

Table -2 : Showing Age distribution of the Sample

Groups	Mean	SD	T	Sig.
Combat	16.1584	2.11534	.550	.099
Non-Combat	16.0000	1.60357		

**Significant at .01 level *Significant at .05 level

These tables shows that the sample was matched with respect to Age, gender, education, religion, family income, residential back-ground, family structure, father's education, mother's education, and language. Some other confounding variable might have been there but the effort was made to keep them under control by keeping the conditions of testing same for all the participants.

Table -3 : Showing Correlations among the Study Variables

	Cognitive flexibility X Psychological performance	Emotional intelligence X Psychological performance	Cognitive Flexibility X Emotional intelligence
Whole Sample	0.314**	0.092	0.080
Combat	0.317**	0.081	0.027
Non-combat	0.382**	0.141	0.433**

**Significant at .01 level

Table 3 shows statistics correlation for combat group, non-combat group and whole sample between cognitive flexibility, emotional intelligence, and psychological performance. According to this Table, a statistically significant correlation (R=0.314) of 0.01 level was found between cognitive flexibility and psychological performance, for the overall sample. It signifies that the players who are cognitively more flexible can very easily find out the negative side of their opponents, the method to get rid of any difficult situation; or the best skill to use during the time of competition. Having good cognitive flexibility makes it easy to visualize all of this.

According to this Table 3, a statistically significant correlation (R=0.317) of 0.01 level was also found between cognitive flexibly and psychological performance for the combat group. Another statistically significant correlation (R=0.382) of 0.01 level was found between cognitive flexibility and psychological performance for the non-combat group too. Players with good Cognitive flexibility keep a keen eye at the crucial assessment of opposite side players, as well as innovative methods for their team to prepare for how difficult and unexpected situations are dealt with. Players with good cognitive flexibility can focus their attention in their game and manage their motivation by having a clear imagery. Players with good cognitive flexibility have full faith in their decisions and strategy and this way they can use their energy in a positive direction easily.

Further, a statistically significant correlation (R= 0.433) of 0.01 level was found between cognitive flexibility and emotional intelligence for the non-combat group. It signifies that the players who are emotionally intelligent remain aware of the soft signals of emotional states and manage their own and other's emotions under varying circumstances easily. Whenever there is any problem or in an unexpected new situation, an emotionally intelligent player tends to manage his own emotions by being in need for an alternative to deal with the difficult situation.

It is assumed that cognitive and emotional constructs are connected. It is particularly striking that cognitive flexibility, which is defined by cognitive psychologist and evaluated as the ability to adapt to certain situations and passing from one thought to another; or the capacity of looking at different problems with multilateral strategies (Stevens, 2009) shows conceptual similarity with emotional intelligence. It is quite difficult to perceive what percentages of exhibited behavior are affected by cognitive or emotional intelligence. However, the present study examines to what extent relationship between emotional intelligence and cognitive flexibility and their contribution are effective in the determining psychological performance, among combat and non-combat sports.

There are a limited number of studies analysing the combined effect of emotional intelligence and cognitive flexibility, in combat and non-combat sports. Both Emotional intelligence and cognitive flexibility, separately, have been discussed in terms of different variables in some studies which have been discussed above. Because of the limited number of studies on the topic, the role of emotional intelligence and cognitive flexibility in determining psychological performance in combat and non-combat sports is not clearly defined.

This study discussed the role of emotional intelligence and cognitive flexibility in determining psychological performance in combat and non-combat sports and aims to address the current gap in knowledge within this domain. It was hypothesized that examining the role of emotional intelligence and cognitive flexibility in determining psychological performance is crucial for performance of combat and non-combat sports. The findings of the limited number of studies examining emotional intelligence and cognitive abilities together are consistent with the findings of the present study.

REFERENCES

Abrenthy, B., Baker, J. & Cote, J. (2005). Transfer of pattern Recall Skills may contribute to the development of Sports expertise. *Applied cognitive psychology,* 19 (6), 705-718.

Anderson, P. (2002). Assessment and Development of Excutive Function (EF) During Childhood. *A Journal on Normal and abnormal development in childhood and adolescence, 71-82.*

Bilgin, M., & Bilissel, E. (2009). Developing a Cognitive Flexibility Scale: Validity and Reliability Studies. *Cukurova Universitesi Egitim Fakultesi Dergisi*, 36 (3), 142-157.

Bulent, G. (2013). Emotional intelligence, cognitive flexibility and psychological symptoms in pre-service teachers. *Global journal of psychology and behavioral education*, 8(13), 1048-1056.

Crombie, D., Lombard, C. & Noakes, T. (2009). Emotional Intelligence Scores predict Team Sports Performance in a National Cricket Competition. *International Journal of Sport Science and Coaching, 4 (2), 209-22.*

Davis, S. K. & Humphrey, N. (2012). Emotional intelligence predicts adolescent Mental Health beyond Personality and Cognitive Ability. *Pers. Individ. Dif.*, 52(2), 144 -149.

Dennis, J. P. & Vander Wal, J. S. (2009a). The Cognitive Flexibility Inventory: Instrument development and estimate of Reliability and Validity. *Journal of Springer Science,* 31, 233-251.

Dennis, J. P., & Vander Wal, J. S. (2009). A comparison of the role of performance-based and self-report measures of Cognitive Flexibility in predicting Depression. *Manuscript in preparation*, Saint Louis University, MO: USA.

Dennis, J. P. & Vander Wal, J. S. (2009). The relationship between life event stress Cognitive Flexibility Coping, and Depression: A Longitudinal Study. *Manuscript in preparation*, Saint Louis University, MO: USA.

Dennis, J. P. & Vander Wal, J. S. (2010b). The Cognitive Flexibility Inventory: Instrument Development and Estimates of Reliability and Validity. *Cogn Ther Res*, 34, 241- 253.

Golby, J., Sheard, M. & Werch, A. (2007). *Evaluating the factor structure of the psychological performance inventory. Sage journals, 105, 309-325.*

Hardy, L., Jones. G. & Gould, D (1996). *Understanding psychological preparation for sport: Theory and practice of elite performers.* John Willey & Sons Inc, Nj: USA.

Lane, A. M., Thelwell, R. C., Lowther, J.P. & Devonport, T. J. (2009). Relationships between Emotional Intelligence and Psychological Skills among Athletes. *Social Behavior and Personality*, 37, 195-202.

Lazarus, S. R. (2000). How Emotions Influence performance In Completive Sports. *Human Kinetic Journlas.* 14 (3) 229-252.

Loehr, J. E. (1986). *Mental toughness training for sports: Achieving athletic excellence.* Stephen Green Press, Lexington: MA.

Mellalieu, D.S., Neil, R. & Fletcher, D. (2009). Competition Stress in Sport performance: Stressors experienced in the Competition environment. *Journal of Sports Sciences,* 27(7), 729-744.

Memmert, D., Baker, J. & Bertsch, C. (2010). Play and practice in the development of Sport-Specific Creativity in Team Ball Sports. *High Ability Studies*, 21(1), 3–18.

Ozturk, M. O. (2004). Emotional Functioning in Depressed and Depression-vulnerable College Students. *Cogn. Emotion,* 17 (5), 799-806.

Pandey, R. & Anand, T. (2003). Factor Structure of a measure of Emotional Intelligence scale in Hindi speaking Indian adolescents. *Indian journal of psychological issues, 11 (1).*

Perlini, H. & Halverson, R.T. (2006). Emotional Intelligence in the National Hockey Team. *Journal of beh. Sci.,* 38 (2), 109-119.

Ranzijn, R. (2005). Does Emotional Intelligence predict unique variance in Life Satisfaction beyond IQ and Personality? *Personality and Individual Differences*, 38 (6), 1353-1364.

Sahin, N. H., Guler, M. & Basim, H. N. (2009). A tipi kisilik oruntusunde bilissel ve duygusal zekanin *stresle basa cıkma ve stres belirtil*eri ile iliskisi. *Turk Psikiyatri Dergisi*, 20 (3), 243-254.

Sit, H.P. & Lindner, J. K. (2005). Motivational Orientations in youth Sport participation: Using Achievement Goal Theory and Reversal Theory. *Personality and individual differences, 38, (3) 605-618.*

Stevens, A. D. (2009). Social Problem-Solving and Cognitive Flexibility: Relations to Social Skills and Problem Behavior of at-risk Young Children. *Unpublished Doctorate Thesis*, Seattle Pacific University.

Thelwell, R. C., Lane, A.M., Wetson, N. J.V. & Greenlees, A. L. (2009). Examining Relationships between Emotional Intelligence and coaching efficacy. *International journal of sport and Exercise psychology*, 6 (2), 224-235.

Zizzi, J., Deaner, H. R. & Hirschhorn, D. K. (2003). The Relationship between Emotional Intelligence Performances among College Baseball Player. *Journal of Applied Sport Psychology*, 15 (3), 262-269.

– o –